

CYBERSECURITY HOME LAB SETUP REPORT

Submitted By

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1. Objective

The objective of this task was to build a safe and isolated cybersecurity practice environment for learning web application security testing, vulnerability assessment, and network traffic analysis.

The lab environment consists of:

- Kali Linux Virtual Machine
- OWASP Juice Shop Vulnerable Application
- Docker Container Environment
- Burp Suite
- Wireshark
- Nmap

2. Software and Tools Used

Tool	Purpose
VirtualBox	Virtualization Platform

Kali Linux	Attacker Machine
Docker	Container Management
OWASP Juice Shop	Vulnerable Web Application
Nmap	Network Scanning
Burp Suite	Web Proxy and Security Testing
Wireshark	Network Packet Analysis

3. Lab Architecture

The lab environment was configured using Kali Linux running inside VirtualBox. OWASP Juice Shop was deployed inside a Docker container hosted on Kali Linux.

Architecture:

```

Internet
|
VirtualBox
|
Kali Linux VM
|
Docker
|
OWASP Juice Shop (Port 3000)

```

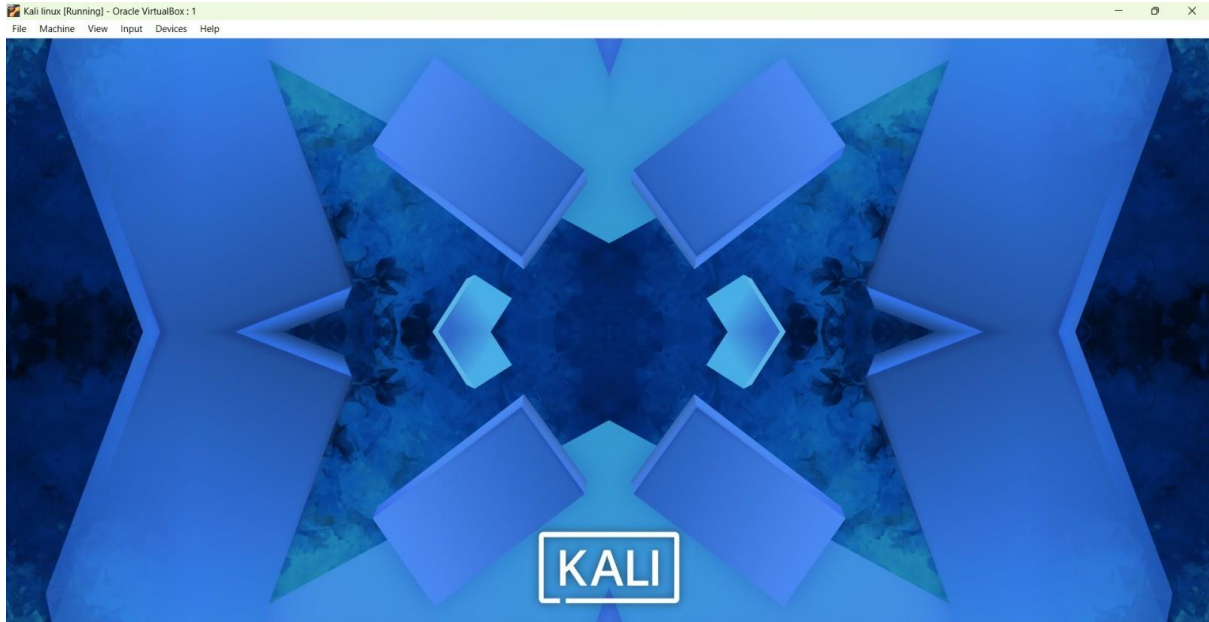
4. Kali Linux Setup

A Kali Linux virtual machine was configured in VirtualBox.

Configuration:

- Adapter 1: NAT
- Docker Service Enabled
- Security Tools Installed

Screenshot:



5. Deployment of OWASP Juice Shop

OWASP Juice Shop was deployed using Docker.

Commands Used:

```
sudo docker start juice
```

The application was successfully started and accessed through:

<http://127.0.0.1:3000>

Screenshot:

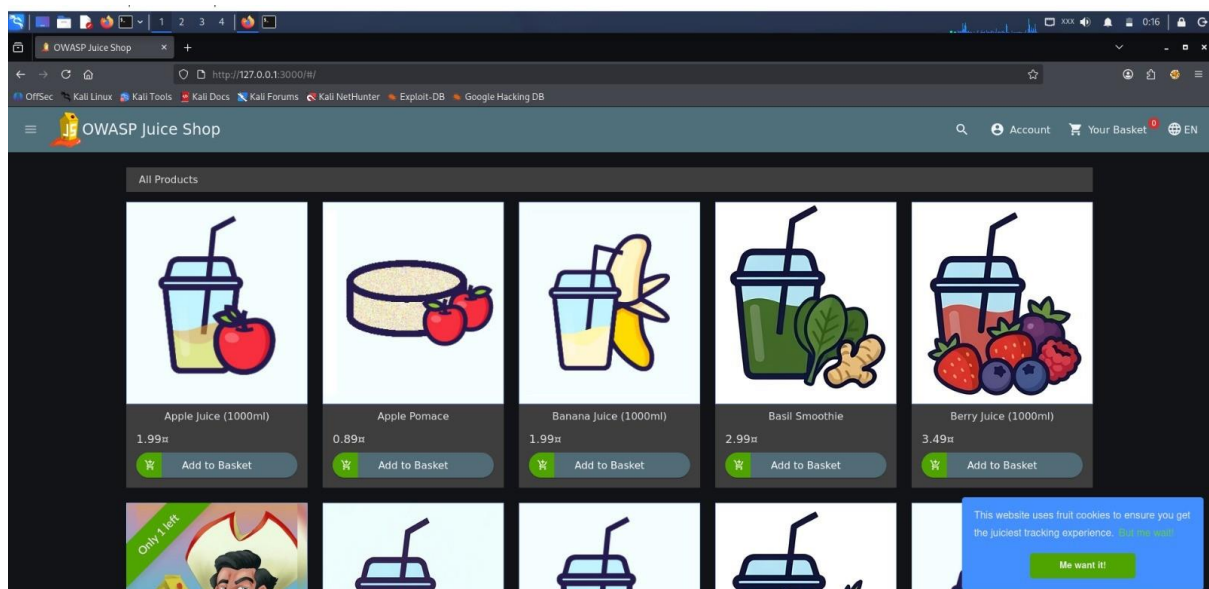
```
Kali linux [Running] - Oracle VM VirtualBox: 1
File Machine View Input Devices Help

ashiga@kali: ~
zsh: corrupt history file /home/ashiga/.zsh_history
(ashiga@kali)~$ sudo docker ps
[sudo] password for ashiga:
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS                    NAMES
7978c00ae312   bkimminich/juice-shop   "/nodejs/bin/node /j_-"   3 minutes ago   Exited (255)   About a minute ago       0.0.0.0:3000->3000/tcp, [::]:3000->3000/tcp   juice

(ashiga@kali)~$ sudo docker start juice
juice

(ashiga@kali)~$ sudo docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS                    NAMES
7978c00ae312   bkimminich/juice-shop   "/nodejs/bin/node /j_-"   5 minutes ago   Up 56 seconds   0.0.0.0:3000->3000/tcp, [::]:3000->3000/tcp   juice

(ashiga@kali)~$
```



6. IP Configuration

The network configuration of the Kali Linux virtual machine was verified using the following command:

```
ip addr
```

Screenshot:

```
(ashiga@kali)-[~]
$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:ca:7e:1a brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0
        valid_lft 85237sec preferred_lft 85237sec
    inet6 fd17:625c:f037:2:a58d:28c4:9b9a:f086/64 scope global temporary dynamic
        valid_lft 86278sec preferred_lft 14278sec
    inet6 fd17:625c:f037:2:a00:27ff:feca:7e1a/64 scope global dynamic mngtmpaddr noprefixroute
        valid_lft 86278sec preferred_lft 14278sec
    inet6 fe80::a00:27ff:feca:7e1a/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
3: docker0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default
    link/ether 92:e8:09:24:52:21 brd ff:ff:ff:ff:ff:ff
    inet 172.17.0.1/16 brd 172.17.255.255 scope global docker0
        valid_lft forever preferred_lft forever
    inet6 fe80::90e0:9ff:fe24:5221/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
4: veth5813431@if2: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue master docker0 state UP group default
    link/ether 26:2b:e5:aa:13:bf brd ff:ff:ff:ff:ff:ff link-netnsid 0
    inet6 fe80::242b:e5ff:feaa:13bf/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
```

7. Network Validation Using Nmap

The Nmap tool was used to identify active services and open ports.

Command Used:

`nmap -sV localhost`

Result:

- Port 3000 detected
- OWASP Juice Shop service reachable

Screenshot:

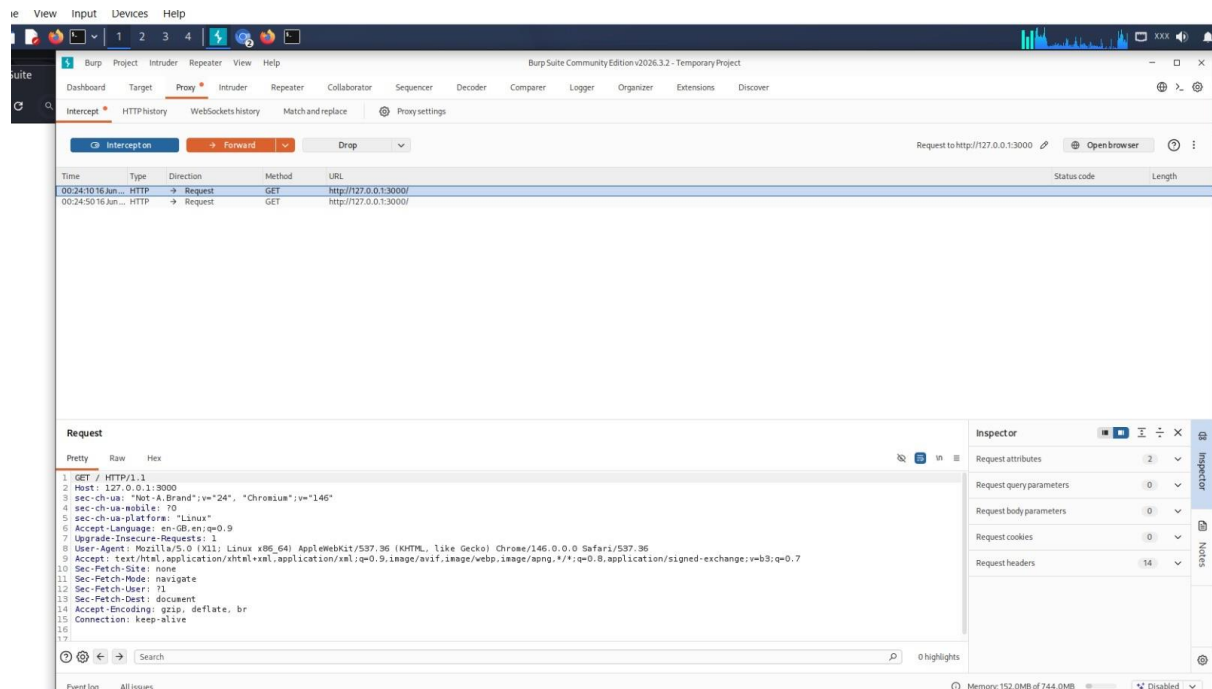
```
Session Actions Edit View Help
→ valid_lft forever preferred_lft forever
(ashiga@kali)-[~]
$ nmap -sV localhost
Starting Nmap 7.99 ( https://nmap.org ) at 2026-06-16 00:18 +0530
Nmap scan report for localhost (127.0.0.1)
Host is up (0.000000s latency).
Other addresses for localhost (not scanned): ::1
Not shown: 997 closed tcp ports (reset)
PORT      STATE SERVICE
443/tcp   open  ssl/https
3000/tcp   open  ppp?
9200/tcp   open  ssl/http Amazon OpenSearch REST API (Basic auth)
2 services unrecognized despite returning data. If you know the service/version, please submit the following fingerprints at https://nmap.org/cgi-bin/submit.cgi?new-service :
=====
NEXT SERVICE FINGERPRINT (SUBMIT INDIVIDUALLY)
SF-Port443-TCP:V=7.99NT=SSLNI=7ND=6/16NTIME=6A30490ANP=x86_64-pc-linux-gnu
SF:rx(GetRequest,154,"HTTP/1.1,x20302(x20Found\r\nlocation:\x20/app/Login
SF:\x\nosd-name:\x20kali\r\nx-frame-options:\x20sameorigin\r\nx-cache-co
SF:rol:\x20private,\x20no-cache,\x20no-store,\x20must-revalidate\r\nset-co
SF:okie:\x20security_authentication:\x20Max-Age=0;\x20Expires=Thu,\x2001\
SF:\x20Jan\x201970\x2000:00;\x20GMT);\x20Secure;\x20HttpOnly;\x20Path=/\
SF:\x20content-length:\x200\r\nDate:\x20Mon,\x2015\x20Jun\x202026\x2018:48:42
SF:\x20GMT\r\nConnection:\x20close\r\n\r\n")\r(HTTPOptions,138,"HTTP/1.1\
SF:\x20404\x20Not\x20Found\r\nosd-name:\x20kali\r\nx-frame-options:\x20same
SF:origin\r\ncontent-type:\x20application/json;\x20charset=utf-8\r\nx-cache-
SF:control:\x20private,\x20no-cache,\x20no-store,\x20must-revalidate\r\nco
SF:\x20content-length:\x2006\r\nDate:\x20Mon,\x2015\x20Jun\x202026\x2018:48:42\r
SF:\x20GMT\r\nConnection:\x20close\r\n\r\n")\r(statusCode=404,"Error":{},"No
SF:\x20Found","message":"\x20Not\x20Found"}")\r(FourOhFourRequest,1C1,"H
SF:TP/1.1\x20404\x20Unauthorized\r\nosd-name:\x20kali\r\nx-frame-options
SF:\x20sameorigin\r\ncontent-type:\x20application/json;\x20charset=utf-8\r
SF:\x20x-cache-control:\x20private,\x20no-cache,\x20no-store,\x20must-revali
SF:rate\r\nset-cookie:\x20security_authentication:\x20Max-Age=0;\x20Expire
SF:=Thu,\x2001\x20Jan\x201970\x2000:00:00;\x20GMT);\x20Secure;\x20HttpOnly;
SF:\x20Path=/\r\ncontent-length:\x2077\r\nDate:\x20Mon,\x2015\x20Jun\x2020
SF:\x2026\x2018:48:42\r\n\r\n")\r(statusCode=404,"Error":{},"message":"\x20Unauthorized\x20Required"
SF:")\r(RFSRequest,138,"HTTP/1.1\x20404\x20Not\x20Found\r\nosd-name:\x2
SF:\x20kali\r\nx-frame-options:\x20sameorigin\r\ncontent-type:\x20application
SF:\x20json;\x20charset=utf-8\r\nx-cache-control:\x20private,\x20no-cache,\x20
```

8. Burp Suite Validation

Burp Suite was launched and configured using FoxyProxy.

The Juice Shop application traffic was successfully intercepted and recorded in HTTP History.

Screenshot:



9. Wireshark Validation

Wireshark was used to capture and analyze network traffic generated during testing.

Captured Traffic:

- HTTP Requests
- Localhost Communication
- Packet Details

The screenshot shows the Wireshark network protocol analyzer interface. The main window displays a list of captured packets. The first packet is selected, and its details are shown in the packet details pane. The status bar at the bottom indicates that 1000 packets were captured and none were dropped.

No.	Time	Source	Destination	Protocol	Length	Info
1	0	Private_01:01:01	Broadcast	IPv4	3950	Bogus IPv4 version (7, 1)
2	1	Private_01:01:01	Broadcast	IPv4	3611	Bogus IPv4 version (9, 1)
3	2	Private_01:01:01	Broadcast	IPv4	4986	Bogus IPv4 version (11, 1)
4	3	Private_01:01:01	Broadcast	IPv4	1990	Bogus IPv4 version (10, 1)
5	4	Private_01:01:01	Broadcast	IPv4	563	Bogus IPv4 version (14, 1)
6	5	Private_01:01:01	Broadcast	IPv4	2422	Bogus IPv4 version (1, 1)
7	6	Private_01:01:01	Broadcast	IPv4	2063	Bogus IPv4 version (10, 1)
8	7	Private_01:01:01	Broadcast	IPv4	937	Bogus IPv4 version (11, 1)
9	8	Private_01:01:01	Broadcast	IPv4	3049	Bogus IPv4 version (8, 1)
10	9	Private_01:01:01	Broadcast	IPv4	982	Bogus IPv4 version (14, 1)
11	10	Private_01:01:01	Broadcast	IPv4	2669	Bogus IPv4 version (1, 1)
12	11	Private_01:01:01	Broadcast	IPv4	886	Bogus IPv4 version (14, 1)
13	12	Private_01:01:01	Broadcast	IPv4	1455	Bogus IPv4 version (0, 1)
14	13	Private_01:01:01	Broadcast	IPv4	1733	Bogus IPv4 version (14, 1)
15	14	Private_01:01:01	Broadcast	IPv4	4937	Bogus IPv4 version (18, 1)

Frame 1: Packet, 3950 bytes on wire (31600 bits), 3 packets captured (0.75%)

- Ethernet II, Src: Private_01:01:01 (01:01:01:01:01:01), Dst: Broadcast
- Internet Protocol Version 4, Src: 192.168.1.1, Dst: 255.255.255.255

Wireshark_Random packet generator\J5RCR3.pcapng | Packets: 1000 - Dropped: 0 (0.0%) | Profile: Default

A cybersecurity home lab environment was successfully established using Kali Linux and OWASP Juice Shop. Docker was used to deploy the vulnerable application, while Nmap, Burp Suite, and Wireshark were utilized to validate connectivity, analyze traffic, and perform security testing activities. The environment provides a safe platform for future web application security testing and cybersecurity learning exercises.